



CS440

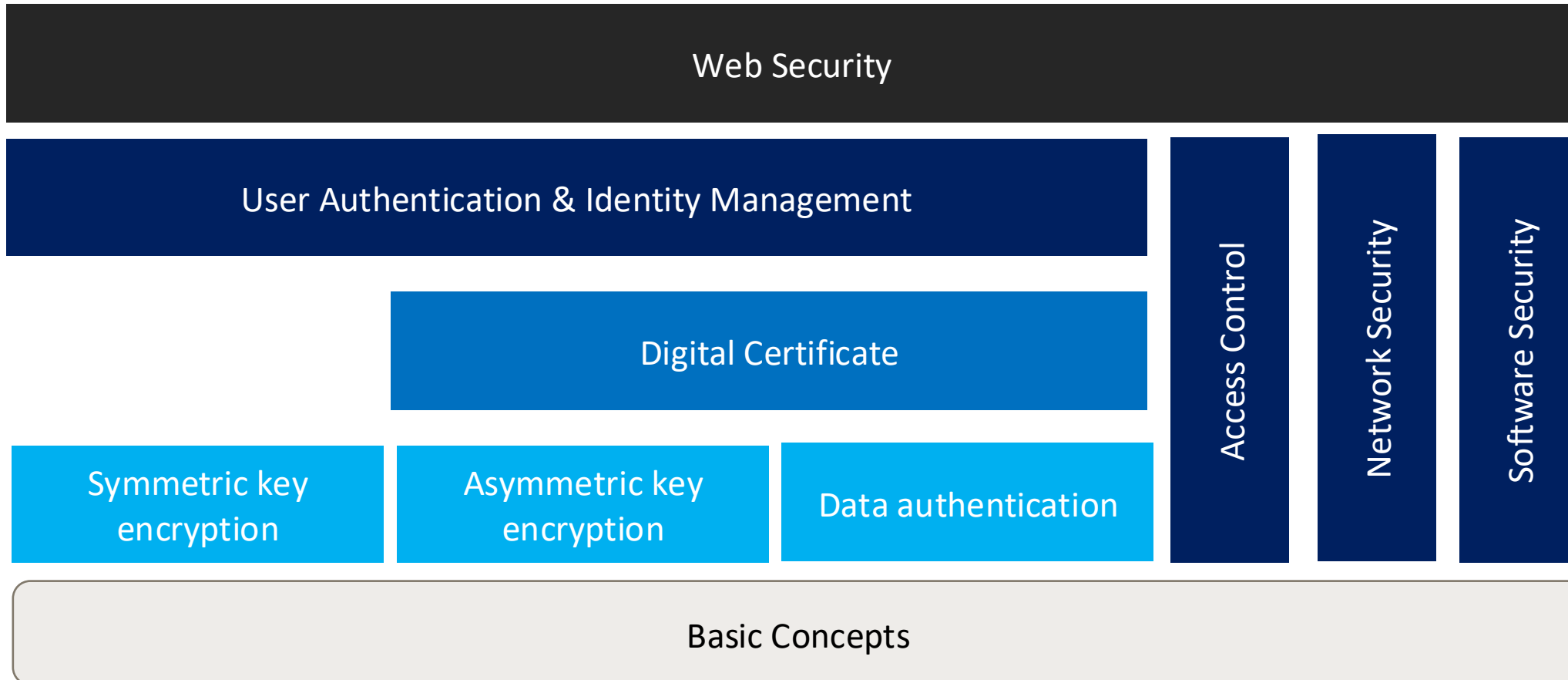
Foundations of Cybersecurity

Introduction

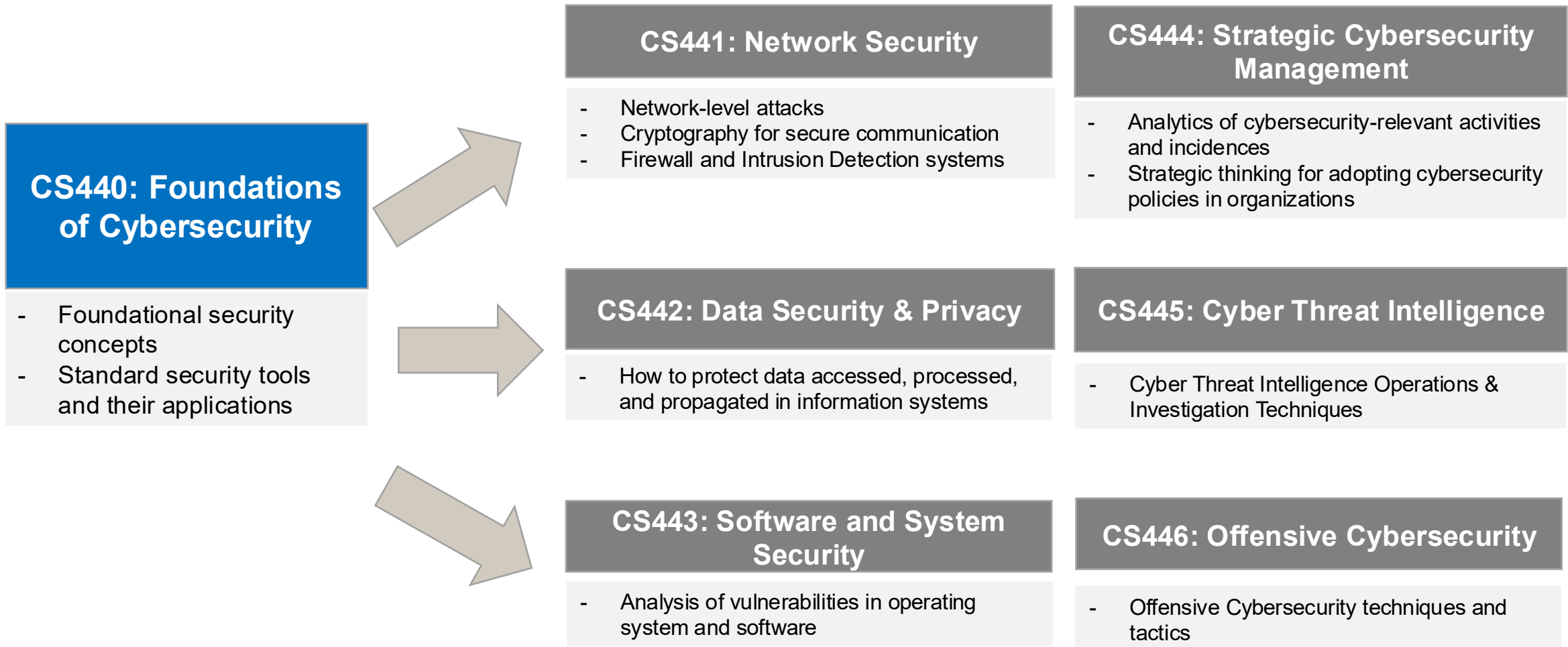
Sections, Lecturers & TAs

Section	Faculty	Date / Time	Venue	TA Name	TA Email
G1	DING Xuhua xhding@smu.edu.sg	Thu / 3:30 pm	SCIS1 Seminar Room 3-3	JIANG Bowen	bowen.jiang.2024@phdcs.smu.edu.sg
G2	DING Xuhua	Fri / 12:00 pm	SCIS1 Seminar Room 3-4	CHENG ChingLam	cl.cheng.2024@phdcs.smu.edu.sg
G3	TAN Kee Hock keehocktan@smu.edu.sg	Wed / 7.00 pm	SCIS1 Seminar Room 3-1	HO Zhao Heng	zh.ho.2024@computing.smu.edu.sg
G4	HUANG Shaofei sfhuang@smu.edu.sg	Thu / 7:00 pm	SCIS1 Seminar Room 2-3	Muskan DHAMURIA	muskan.d.2024@computing.smu.edu.sg

Syllabus



Course Linkages



Course Structure

Week	Topic
01	Basic concepts of cybersecurity
02	Symmetric Key Encryption (Confidentiality)
03	Asymmetric Key Encryption (Confidentiality) -- Project Briefing
04	Data Authentication & Integrity
05	Public Key Certificate
06	User Authentication & Identity Management
07	Access Control
08	Recess (Midterm 1.5hrs (20%))
09	Network Security
10	Software Security
11	Web security I
12	Web security II –
13	Project presentations by student groups (20%)
14	Study week
15	Exam week (45%)

Assessment

Week	Course Assessment	Weightage
01		Continuous Assessment : 55% ▪ Mid-term Test (20%) ▪ Project (20%) ▪ Class Participation (15%)
02		
03		
04		
05		Exam : 45%
06		
07		
08	Mid-term (6 Oct, 19:00 ~ 20:30)	
09		
10		
11		
12		
13	Project presentations by student groups	20%
14	Class participations	15%
15	Exam week	45%

Course Assessment

- Class participation 15%
 - Class attendance, participations, labs, pop quizzes
- Midterm test 20%
 - MCQs and Short Answer Questions
- Project 20%
 - Project presentation (week 13)
- Final exam 45%
 - MCQs and Short Answer Questions
- All the tests are closed-book. Allow non-programmable/non-graphical calculator.

Pop Quizzes

- Pop quizzes in eLearn.
- All MCQs
- 5% in total (within 15% class participation)

Project: Learning Targets

- Understand a security technique.
- Learn how to analyze a problem.
- Learn how to present in a logical way.
- Learn how to learn.

Project: What to Achieve

- Understand the security technology.
 - what is the problem addressed by the technology
 - the setting, the adversary
 - what are the solutions being recommended or in deployment
 - why can they solve the problem? pros and cons of the solutions.

Project: Deliverables

- Presentation

- Submit your slides to eLearn Assignment by the mid-night before your presentation date
- Presentation and Q/A in Week 13 Lecture period
- Time: 12 minutes presentation + 3 minutes Q/A

- Evaluation Criteria

- Quality of presentation (5%)
- Basic understanding of the technology (application, strength, weakness et) (10%)
- Depth of understanding of the problem and technology. (5%)

Project: Teaming

- Form your own team (**by Week 4**)
 - 4 members by default; instructor approval is required for 5-member teams.
- Topic selection (**by Week 5**)
 - choose one topic from the list in the project briefing file.
 - instructor approval is required for topics outside of the list.
 - Teams in the same section are **not** allowed to choose the same topic.
 - email your TA about your project topic selection. Contention is solved according the request's temporal order. First come first serve.
 - Click [here](https://docs.google.com/spreadsheets/d/1f1rU_wfXxKeW4F2avjIKCKt73B-NXth_g8AMbs9SLYQ/edit?usp=sharing) to view the status of topic selection.

(https://docs.google.com/spreadsheets/d/1f1rU_wfXxKeW4F2avjIKCKt73B-NXth_g8AMbs9SLYQ/edit?usp=sharing)

Project: Penalty for Free-riders

- By default, every team member will get the same grade
- If there is a case of non-participation, report to the instructor as early as possible; it will be handled on case-by-case basis
 - Can result in grade differences among team members or even failure
- Be responsible to your team and yourself!

Course Policies: Attendance

- 1 point (of final grade) will be taken off for each absence without a valid reason
- May fail the course if you miss 3 classes
- Inform us early if you need to attend other section with valid reasons
 - Email your instructor with your reasons and get approval in advance
 - Inform both TAs in your original class and the makeup class to transfer your attendance back

Academic Integrity

- All acts of academic dishonesty (including, but not limited to, plagiarism, cheating, fabrication, facilitation of acts of academic dishonesty by others, unauthorized possession of exam questions, or tampering with the academic work of other students) are serious offences.
- All work (whether oral or written) submitted for purposes of assessment must be the student's own work. Penalties for violation of the policy range from zero marks for the component assessment to expulsion, depending on the nature of the offence.
- When in doubt, students should consult the instructors of the course.
- Details on the SMU code of Academic Integrity may be accessed [here!](#)

Plagiarism and Cheating

- When plagiarism and cheating case happens, during the quizzes/midterm/project/exam, this is the process we follow:

Detection

- E-learn team or Faculty
- Inform Dean's office

Investigation

- E-learn team
- Faculty
- Dean's office
- Office of Dean of Students

Sanctions

- Office of Dean of Students
- Dean's office
- RO

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Survival Tips

- Before class
 - Read the slides/reference text before class!
- In class
 - Pay attention
 - Do workshop exercises
 - Clarify doubts and participate
 - Take notes
- After class
 - Revise your notes
 - Ask questions and clarify doubts
 - Participate in online discussions